

SQL Joins Explained

Subject: Database Management Systems Topic: SQL

Overview

Inner, left, right, full outer joins with examples and performance tips

Join fundamentals

- A join combines rows using a predicate, commonly a foreign-key to primary-key equality.
- INNER JOIN returns matching row pairs only.
- LEFT JOIN returns every left row and fills right-side columns with NULL when no match exists.

Join variants

- RIGHT JOIN preserves all right-side rows; swapping table order often makes the same query easier to read as LEFT JOIN.
- FULL OUTER JOIN preserves unmatched rows from both sides where the database supports it.
- CROSS JOIN produces every pair of rows and can grow to the product of both table sizes.

Example and NULL behavior

- `SELECT s.name, e.course_id FROM students s LEFT JOIN enrollments e ON e.student_id = s.id` preserves students with no enrollment.
- A WHERE condition on the right table can remove NULL-extended rows and effectively turn a left join into an inner join.
- Put optional right-side filters in the ON clause when unmatched left rows must remain.

Performance

- Index join keys used for lookups and inspect the query plan before optimizing.
- Filter early when it preserves query meaning, and select only the columns needed.
- Check for duplicate join keys: one left row can match multiple right rows and increase result cardinality.